



Tianhua Gao

Ph.D. Candidate

Personal Website: <https://tianhuagao.github.io/>

| About Me

Tianhua Gao is currently pursuing the Ph.D. degree with the Graduate School of Systems and Information Engineering, **University of Tsukuba**, Japan. He is also a Research Assistant with the Intelligent Systems Research Institute, **National Institute of Advanced Industrial Science and Technology (AIST)**, Japan. His research objective is to integrate control theory and neuroscience to advance the field of machine intelligence & control. He has completed his Ph.D. dissertation defense and is expected to receive the Ph.D. degree in March 2026.

| Academic & Professional Memberships

- IEEE Graduate Student Member
- ASME Graduate Student Member

| Awards

- SPRING Scholarship 2023-2026
¥8,100,000 grant for PhD research. Support for Pioneering Research Initiated by the Next Generation (SPRING).
- University of Tsukuba Fellowship 2023
Full tuition coverage provided during the Ph.D. program.

| Publications

- **T. Gao**, M. Izumita, K. Tomita and A. Kamimura, "A Sliced Learning Framework for Online Disturbance Identification in Quadrotor SO(3) Attitude Control," in *IEEE/ASME Transactions on Mechatronics*, 2026 (Under Review).
- **T. Gao**, K. Tomita and A. Kamimura, "Online Identification using Adaptive Laws and Neural Networks for MultiQuadrotor Centralized Transportation System," *Engineered Science*, 2025.
- **T. Gao**, K. Tomita and A. Kamimura, "Robustness Enhancement for Multi-Quadrotor Centralized Transportation System via Online Tuning and Learning," *American Control Conference (ACC)*, Denver, CO, USA, 2025.
- **T. Gao**, X. Han, K. Tomita and A. Kamimura, "Numerical Simulation of a Novel Cargo Handling Strategy: Using a Centralized Cable-Linked Dual-Multirotor System," *IEEE International Conference on Advanced Intelligent Mechatronics (AIM)*, Boston, MA, USA, 2024, pp. 345-352 .
- **T. Gao**, X. Han, K. Tomita and A. Kamimura, "Nonlinear Model-Assisted Control for Autonomous Parachute Precision Landing Recovery System," *61st Annual Conference of the Society of Instrument and Control Engineers (SICE)*, Kumamoto, Japan, 2022, pp. 1158-1164.

| Technical Background

Mathematics:

- Topology
- Differential geometry and differential manifolds
- Lie groups and Lie algebras
- Graph theory and matrix theory
- Fiber bundle theory

Physics:

- Newtonian mechanics
- Lagrangian mechanics
- Hamiltonian mechanics
- Analytical mechanics

Control Theory:

- Classical control (PID)
- Optimal control (LQR, MPC, NMPC)
- Robust and adaptive control (ADRC, MRAC, L1 adaptive control)
- Nonlinear control and geometric control
- $SO(3)$ -based attitude control and $SE(3)$ coupled control
- Learning-based and neural network control
- Convex and non-convex optimization
- System identification and online learning

Uncrewed Aerial Vehicles (UAVs):

- ArduPilot (APM) and PX4 flight control systems
- Software-in-the-loop (SITL) simulation
- Gazebo simulation
- ROS and ROS 2
- Motion capture (MoCap)-based experimental systems

Programming & Computing:

- C, C++
- Python
- MATLAB / Simulink